

Clinical safety dashboard — Impact of balancing on glucose prediction accuracy

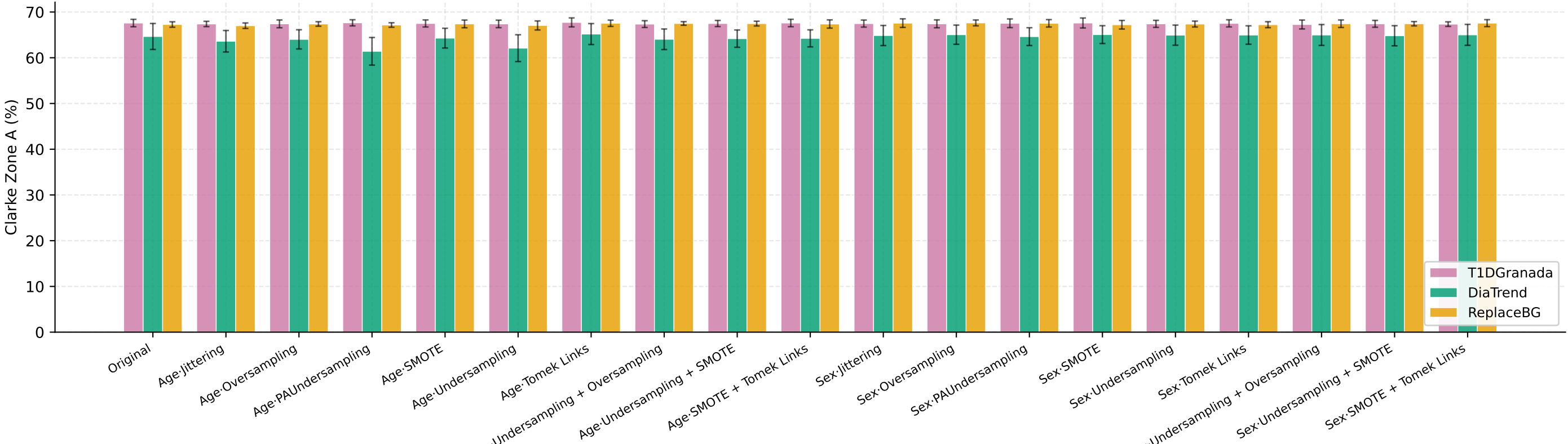
A | RMSE improvement · Hypo L2 (<54 mg/dL) — CRITICAL

	T1DGranada	DiaTrend	ReplaceBG
Age-Jittering	-1.1%	+2.3%	-1.0%
Age-Oversampling	-0.2%	+2.7%	+0.4%
Age-PAUndersampling	-1.1%	-5.8%	-0.6%
Age-SMOTE	+1.6%	+3.4%	+1.2%
Age-Undersampling	-0.3%	-4.9%	-0.3%
Age-Tomek Links	-0.3%	+1.9%	+0.4%
Age-Undersampling + Oversampling	-0.5%	+1.6%	+0.5%
Age-Undersampling + SMOTE	+1.5%	+1.9%	+0.7%
Age-SMOTE + Tomek Links	+0.8%	+2.6%	+0.4%
Sex-Jittering	-1.1%	+0.2%	+1.4%
Sex-Oversampling	-0.7%	+1.2%	+0.6%
Sex-PAUndersampling	+0.1%	-1.1%	+2.1%
Sex-SMOTE	-0.6%	+1.3%	-0.3%
Sex-Undersampling	-0.9%	+0.6%	+0.2%
Sex-Tomek Links	-1.5%	+1.5%	-0.1%
Sex-Undersampling + Oversampling	-1.0%	+1.5%	+0.9%
Sex-Undersampling + SMOTE	+0.1%	+0.5%	+1.4%
Sex-SMOTE + Tomek Links	-0.6%	+0.9%	+0.3%

B | RMSE improvement · Hypo L1 (54-69 mg/dL)

	T1DGranada	DiaTrend	ReplaceBG
Age-Jittering	-0.4%	+2.6%	-1.0%
Age-Oversampling	+0.3%	+3.7%	+0.7%
Age-PAUndersampling	-1.4%	-4.4%	-0.8%
Age-SMOTE	+1.3%	+5.0%	+2.0%
Age-Undersampling	-0.2%	-3.7%	-0.3%
Age-Tomek Links	+0.5%	+1.2%	+0.8%
Age-Undersampling + Oversampling	-0.6%	+2.1%	+0.5%
Age-Undersampling + SMOTE	+0.9%	+3.6%	+1.6%
Age-SMOTE + Tomek Links	+0.8%	+4.0%	+0.6%
Sex-Jittering	-0.1%	+0.7%	+2.1%
Sex-Oversampling	+0.2%	+0.6%	+1.2%
Sex-PAUndersampling	+0.3%	-0.6%	+2.6%
Sex-SMOTE	+0.0%	+1.0%	-0.0%
Sex-Undersampling	-0.6%	+0.9%	+0.5%
Sex-Tomek Links	-1.0%	+1.4%	-0.1%
Sex-Undersampling + Oversampling	-0.6%	+1.1%	+1.2%
Sex-Undersampling + SMOTE	+1.1%	+1.0%	+1.8%
Sex-SMOTE + Tomek Links	-0.3%	+1.3%	+0.5%

C | Clarke Zone A (%) — clinically safe predictions ↑ better



D | RMSE improvement · In Range (70-180 mg/dL)

	T1DGranada	DiaTrend	ReplaceBG
Age-Jittering	-0.3%	+3.9%	+0.1%
Age-Oversampling	+0.2%	+4.4%	+0.0%
Age-PAUndersampling	-0.9%	+1.8%	-0.6%
Age-SMOTE	-0.2%	+3.6%	-0.7%
Age-Undersampling	-0.4%	+0.5%	-1.0%
Age-Tomek Links	-0.5%	+0.7%	-0.4%
Age-Undersampling + Oversampling	+0.2%	+3.6%	-0.5%
Age-Undersampling + SMOTE	-0.3%	+3.1%	-0.5%
Age-SMOTE + Tomek Links	+0.0%	+3.3%	-0.4%
Sex-Jittering	+0.5%	+1.6%	-0.3%
Sex-Oversampling	-0.1%	+1.4%	-0.6%
Sex-PAUndersampling	+0.1%	+1.8%	-0.4%
Sex-SMOTE	-0.2%	+1.4%	-0.1%
Sex-Undersampling	+0.4%	+1.3%	-0.5%
Sex-Tomek Links	-0.9%	+1.2%	-0.4%
Sex-Undersampling + Oversampling	+0.1%	+1.3%	-0.5%
Sex-Undersampling + SMOTE	-0.1%	+1.5%	-0.7%
Sex-SMOTE + Tomek Links	+0.2%	+1.0%	-0.5%

E | RMSE improvement · Hyper L1 (181-250 mg/dL)

	T1DGranada	DiaTrend	ReplaceBG
Age-Jittering	-0.8%	-5.5%	-1.6%
Age-Oversampling	-1.2%	-6.3%	-1.0%
Age-PAUndersampling	+1.4%	-2.8%	-0.5%
Age-SMOTE	-1.0%	-5.7%	-1.0%
Age-Undersampling	-0.2%	-1.9%	-0.2%
Age-Tomek Links	-0.1%	+0.2%	-0.5%
Age-Undersampling + Oversampling	-1.0%	-5.3%	-0.1%
Age-Undersampling + SMOTE	-0.3%	-6.1%	-0.7%
Age-SMOTE + Tomek Links	-0.3%	-5.8%	-1.3%
Sex-Jittering	-0.6%	-1.6%	-0.2%
Sex-Oversampling	-0.5%	-1.3%	-0.1%
Sex-PAUndersampling	-0.6%	-2.0%	-1.0%
Sex-SMOTE	-0.0%	-1.3%	-0.7%
Sex-Undersampling	-0.2%	-1.5%	-0.4%
Sex-Tomek Links	+0.7%	-1.0%	-0.7%
Sex-Undersampling + Oversampling	-0.7%	-0.8%	-0.5%
Sex-Undersampling + SMOTE	-1.3%	-1.7%	-1.0%
Sex-SMOTE + Tomek Links	-0.3%	-1.1%	+0.0%